

laser gyro:

$$t_1 = \frac{2\pi R}{c + \dot{\theta} R} \lambda \frac{2\pi R}{c} \left(1 - \frac{R}{c} \dot{\theta}\right)$$

$$t_2 = \frac{2\pi R}{c - \dot{\theta} R} \lambda \frac{2\pi R}{c} \left(1 + \frac{R}{c} \dot{\theta}\right)$$

$$\Rightarrow \Delta t = t_2 - t_1 = \frac{2\pi R}{c} \cdot \frac{2\pi R}{c} \dot{\theta} = \frac{4\pi R^2}{c^2} \dot{\theta}$$

$$\Rightarrow dL = c \Delta t = \frac{4\pi R^2}{c} \dot{\theta} = \lambda \cdot \frac{d\phi}{2\pi}$$

$$\Rightarrow \dot{\theta} = \frac{c}{4\pi R^2} \cdot \frac{2}{2\pi} d\phi$$

$$\Rightarrow \dot{\theta} = \frac{\lambda c}{8\pi^2 R^2} d\phi$$

